

Agentic Data Pipeline Orchestration for NASA Reanalyses — Colorado Snowpack

MERRA-2 and ERA5, Water Years 1981–2026

Built with Coding Agents and Agent Skills

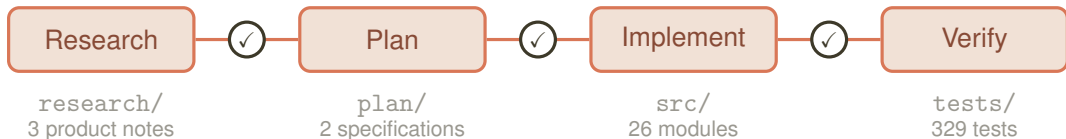
Responsible Gen-AI for NASA Earthdata Hackweek

responsible-genai.hackweek.io

August 28, 2026

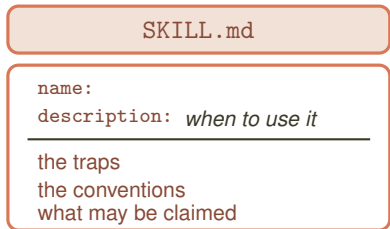


Project Planning



the agent matches on the description

... so it loads itself,
before it can get it wrong



Project Skills

Science

reanalysis-snowpack-comparison
snow-hydrology-fsca-evaluation
modis-merra-regridding
snow-bias-statistics-and-figures

MERRA-2 against ERA5, and which claim holds the Colorado fSCA comparison, end to end 500 m pixels onto coarse model cells bias, MAE, significance, low-snow masks

Run Integrity

earthdata-streaming-checkpoints
resumable-fetch-run-integrity
derived-quantity-integrity

thousands of granules, atomic checkpoints
a run interrupted, resumed, or partial
a ratio or rate over aggregated data

Communication

designing-thematic-figures
building-beamer-decks

one point per figure, nothing spare
this deck: Tectonic, theme seam, gates

Nine skills, written during the work, from what went wrong in it.

The Data Pipeline

MERRA-2 Cloud OPeNDAP • earthaccess

ERA5 ARCO-ERA5 Zarr • Google Cloud

MODSCAG NSIDC DAAC • Earthdata Login

SNOTEL NRCS AWDB REST • WTEQ

3DEP USGS elevation • staged once



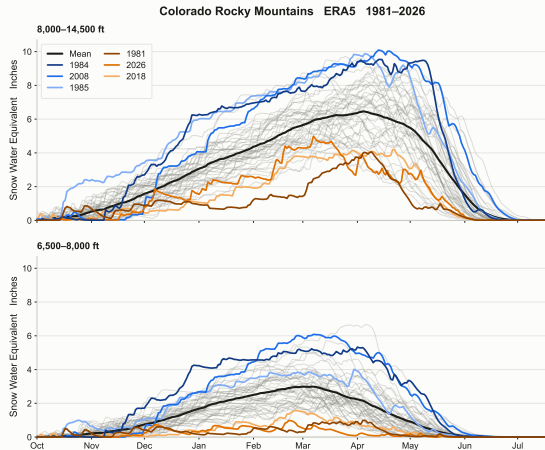
*resumable, and the whole report rebuilds
from the checkpoints with no network*

One Silent Trap per Archive

- **ERA5** *sd* is snow *water equivalent*, not depth. Geometric depth is sd / rsn
- **MERRA-2** *SNODP* is depth *within* the snow-covered fraction; the grid mean is $SNODP \times FRSNO$
- **ARCO-ERA5** the time axis runs to 2050. Absent chunks read back as NaN, with no error
- **SNOTEL** every station sits above the domain's median cell. Timing only, never magnitude

Each returns a plausible wrong number rather than an error. Each is now written down in a skill.

ERA5 by Elevation Band



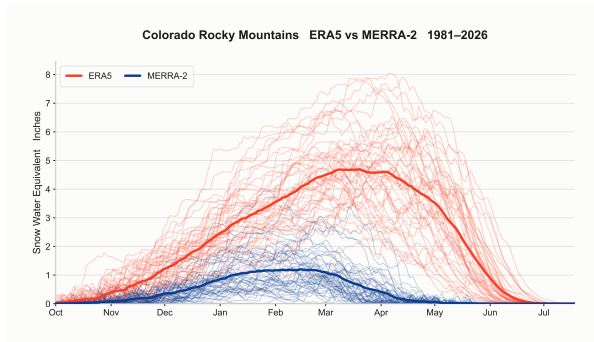
1 April SWE, per cent of band mean

	2023	2026
8,000–14,500 ft	140%	51%
6,500–8,000 ft	297%	5%

WY2026 is the lowest 1 April SWE of the 46 years, at 41% of the record mean. The deficit is not uniform: the lower band all but vanished.

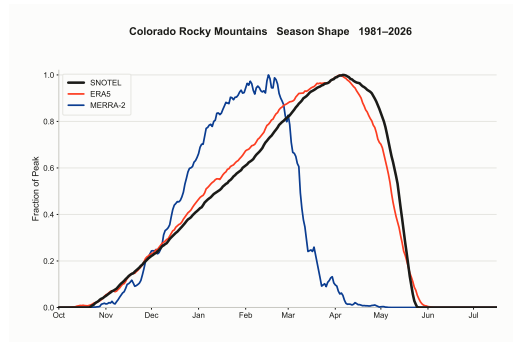
All values ERA5. MERRA-2 gives different amounts for the same snow.

Two Reanalyses, One Observational Check



A factor of 3.3 apart in mean peak SWE.
MERRA-2 exceeds ERA5's mean in 0 of 46 years.

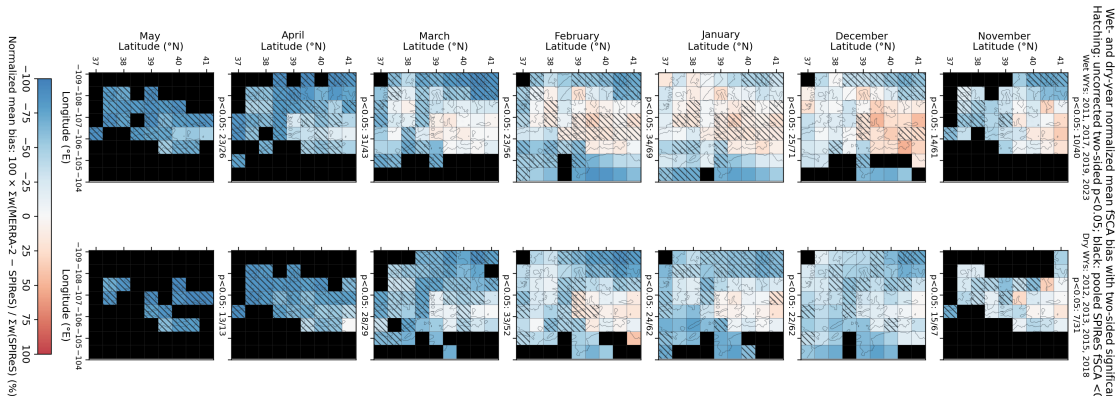
Yet they rank the years alike: $\rho = 0.73$



Each series scaled by its own peak.
SNOTEL arbitrates timing only.

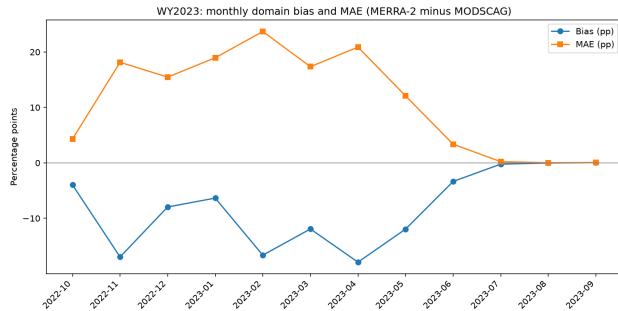
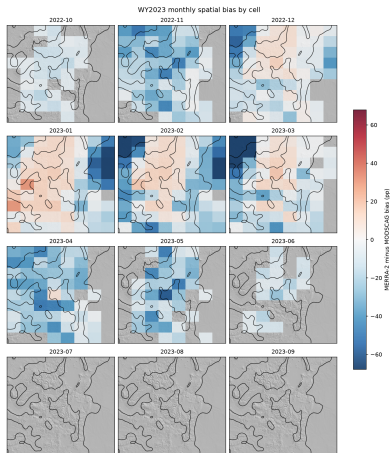
MERRA-2 peaks two months early

fSCA Bias, Wet Years against Dry



MERRA-2 minus SPIReS, normalized mean fSCA bias, WY2010–2023. Wet water years above, dry below; May at left, November at right. Hatching marks uncorrected two-sided $p < 0.05$.

fSCA Bias Month by Month, Water Year 2023

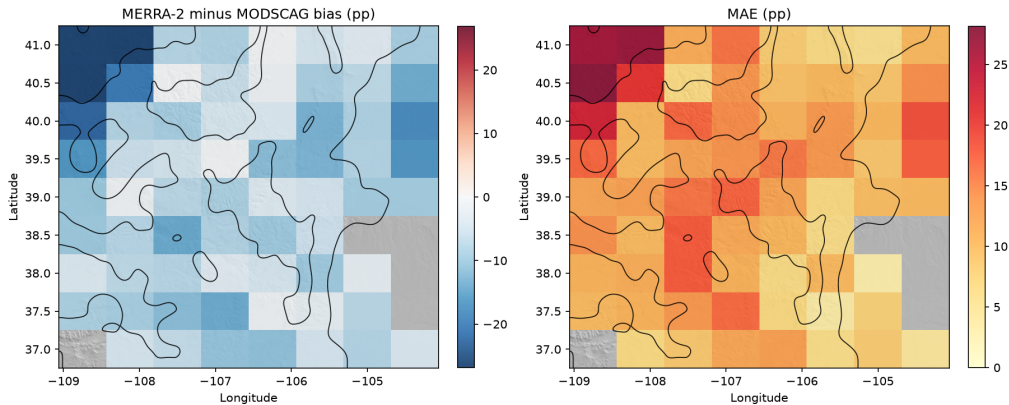


MERRA-2 minus MODSCAG

paired days 347 of 365
mean bias -4.3 pp
MAE 5.7 pp

The model runs low against the satellite through accumulation and converges at melt-out.

The Water Year as a Whole, Water Year 2023



Mean bias and MAE by cell, 347 paired days. Both largest in the northwest.

Melt-out, the last day cover stays above 0.10 — MODSCAG 20 May, ERA5 8 June, MERRA-2 26 April